The Ramsey–Cass–Koopmans Model

Developed by Ramsey (1928), Cass (1965), and Koopmans (1965).

Note: This content will help you once you have gone through the book. If you did not read this topic from the book before, you may face some trouble. In that case, check that particular topic from the book or simply ask someone to explain the issue who have the habit of reading books. Also do not use any unfair means in the exam using this content (or any content).

2.1 Assumptions

Firms

- There are a large number of identical firms
- Each has access to the production function ***Y = F (K,AL)***
- The firms hire workers and rent capital in competitive factor markets
- Firms sell their output in a competitive output market
- Firms take A as given
- A grows exogenously at rate g.
- The firms maximize profits
- Firms are owned by the households, so any profits they earn accrue to the households

Households

- There are large number of identical households
- The size of each household grows at rate n
- The household supplies labor (each member supplies 1 unit labor) and rents capital to firms
- It has initial capital holdings of K(0)/H
  Where, K(0) = amount of capital in the economy
  H = number of households
- K(0) is strictly positive
- There is **no depreciation**
- The household divides its income at each point in time between consumption and saving so as to maximize its lifetime utility
- The household earns from the labor and capital it supplies and, potentially, from the profits it receives from firms
- The household's utility function

$$U = \int_0^{\infty} e^{-\rho t} u(C(t)) \frac{L(t)}{H} dt \tag{2.1}$$

C(t) = the consumption of each member of the household at time t
u(•) = the instantaneous utility function, which gives each member's utility at a given date.
L(t) = the total population of the economy;
L(t)/H = the number of members of the household.
ρ = the discount rate;
The greater is ρ, the less the household values future consumption relative to current consumption.

- The instantaneous utility function takes the form

$$u(C(t)) = \frac{C(t)^{1-\theta}}{1-\theta} \qquad \text{with } \theta > 0 \text{ and } \rho - n - (1-\theta)g > 0 \tag{2.2}$$

It is known as constant-relative-risk-aversion (or CRRA)

(Following part can be skipped if you want)

- θ = coefficient of relative risk aversion for this CRRA utility function (independent of c)
- θ = determines the household's willingness to shift consumption between different periods.
- When θ is smaller, the household is more willing to allow its consumption to vary over time.
- If θ is close to zero, utility is almost linear in C, and so the household is willing to accept large swings in consumption to take advantage of small differences between the discount rate and the rate of return on saving.

(Above part can be skipped if you want)

Three features of the instantaneous utility function:

First, $C^{1-\theta}$ is increasing in C if $\theta < 1$ but decreasing if $\theta > 1$
Second, in the special case of $\theta \to 1$, the instantaneous utility function simplifies to lnC
Third, the assumption that $\rho - n - (1 - \theta)g > 0$ ensures that lifetime utility does not diverge

2.2 The Behavior of Households and Firms

Firms

- Firms employ labor and capital, pay them their marginal products, sell the output.
- Because the production function has constant returns and the economy is competitive, firms earn zero profits.
- The MP_K is $f'(k)$,
- Because markets are competitive, capital earns its marginal product.
- And because there is no depreciation, the real rate of return on capital equals its earnings per unit time.
- Thus the real interest rate at time t is

$$\boldsymbol{r(t) = f'(k(t))} \tag{2.3}$$

- MP_L is $\frac{\partial F(K,AL)}{\partial L} = A\frac{\partial F(K,AL)}{\partial (AL)}$
- In terms of f (•), this is $A[f(k) - kf'(k)]$
- Thus the real wage at time t is

$$W(t) = A(t)[f(k(t)) - k(t)f'(k(t))] \tag{2.4}$$

So, $\frac{W(t)}{A(t)} = w(t) = [f(k(t)) - k(t)f'(k(t))]$

- The wage per unit of effective labor is therefore

$$\boldsymbol{w(t) = f(k(t)) - k(t)f'(k(t))} \tag{2.5}$$

Households' Budget Constraint

- The representative household takes the paths of r and w as given.
- Its budget constraint is that the present value of its lifetime consumption cannot exceed its initial wealth plus the present value of its lifetime labor income.
- As r may vary over time, we define $R(t) = \int_{\tau=0}^{t} r(\tau) d\tau$
- **time 0** — 1 unit good (invested) $\Longrightarrow$ **time t** — $e^{R(t)}$
- Time 0 — $\boldsymbol{e^{-R(t)}}$ $\Longleftarrow$ **time t** — 1 unit output

- One unit of the output good invested at time 0 yields $e^{R(t)}$ units of the good at t
- the value of 1 unit of output at time t in terms of output at time 0 is $\boldsymbol{e^{-R(t)}}$

Now,

- Hosehold Member: $\frac{L(t)}{H}$
- Hosehold's labor income at t: $\frac{W(t)L(t)}{H}$
- Consumption Expenditure: $\frac{C(t)L(t)}{H}$
- Initial Wealth: $\frac{K(0)}{H}$
- The household's budget constraint is therefore

$$\int_{t=0}^{\infty} e^{-R(t)}\boldsymbol{C(t)}\frac{L(t)}{H}\, dt \le \frac{\boldsymbol{K(0)}}{H} + \int_{t=0}^{\infty} e^{-R(t)}\boldsymbol{W(t)}\frac{L(t)}{H}\, dt \tag{2.6}$$

(2.6) can be rewritten as:

$$\frac{\boldsymbol{K(0)}}{H} + \int_{t=0}^{\infty} e^{-R(t)}[\boldsymbol{W(t)} - \boldsymbol{C(t)}]\frac{L(t)}{H}\, dt \ge 0 \tag{2.7}$$

We can write the integral from t = 0 to t = ∞ as a limit. Thus (2.7) is equivalent to

$$\lim_{s\to\infty}\left[\frac{K(0)}{H} + \int_{t=0}^{s} e^{-R(t)}[W(t) - C(t)]\frac{L(t)}{H}\, dt\right] \ge 0 \tag{2.8}$$

Now note that the household's capital holdings at time s are

$$\frac{K(s)}{H} = \boldsymbol{e^{R(s)}}\frac{\boldsymbol{K(0)}}{H} + \int_{t=0}^{s} \boldsymbol{e^{R(s)-R(t)}}[\boldsymbol{W(t)} - \boldsymbol{C(t)}]\frac{L(t)}{H}\, dt$$

$$\Rightarrow \frac{K(s)}{H} = e^{R(s)}\left[\frac{K(0)}{H} + \int_{t=0}^{s} e^{-R(t)}[W(t) - C(t)]\frac{L(t)}{H}\, dt\right] \quad (2.9)$$

The expression in (2.9) is $e^{R(s)}$ times the expression in brackets in (2.8).

Thus, we can write the budget constraint as simply:

$$\lim_{s\to\infty} e^{-R(s)} \frac{K(s)}{H} \geq 0 \quad (2.10)$$

- Expressed in this form, the budget constraint states that the **present value of the household's asset holdings cannot be negative** in the limit.
- Equation (2.10) is known as the **no-Ponzi-game condition**.
- A Ponzi game is a scheme in which someone issues debt and rolls it over forever.
- By imposing the budget constraint (2.6) or (2.10), we rule out such schemes.

Households' Maximization Problem

- The representative household wants to maximize its **lifetime utility** subject to its budget constraint.
- it is easier to work with variables normalized by the **quantity of effective labor**.

We start with the **objective function:**
Define c(t) to be consumption per unit of effective labor
Where, $\mathbf{c(t)} = \frac{C(t)}{A}$
Thus, Consumption per Worker, $C(t) = A(t)c(t)$

The household's instantaneous utility, (2.2), is therefore

$$\frac{C(t)^{1-\theta}}{1-\theta} = \frac{[A(t)c(t)]^{1-\theta}}{1-\theta}$$

$$= \frac{[A(0)e^{gt}]^{1-\theta} c(t)^{1-\theta}}{1-\theta}$$

$$= A(0)^{1-\theta} e^{(1-\theta)gt} \frac{c(t)^{1-\theta}}{1-\theta} \quad (2.11)$$

Substituting (2.11) and the fact that $L(t) = L(0)e^{nt}$ into the household's objective function, (2.1)–(2.2), yields

$$U = \int_{t=0}^{\infty} e^{-\rho t} \frac{C(t)^{1-\theta}}{1-\theta} \frac{L(t)}{H} dt$$

$$= \int_{t=0}^{\infty} e^{-\rho t} \left[\boldsymbol{A(0)^{1-\theta}} e^{(1-\theta)gt} \frac{c(t)^{1-\theta}}{1-\theta} \right] \frac{\boldsymbol{L(0)} e^{nt}}{\boldsymbol{H}} dt$$

$$= \boldsymbol{A(0)^{1-\theta}} \frac{\boldsymbol{L(0)}}{\boldsymbol{H}} \int_{t=0}^{\infty} e^{-\rho t} e^{(1-\theta)gt} e^{nt} \frac{c(t)^{1-\theta}}{1-\theta} dt$$

$$= \boldsymbol{A(0)^{1-\theta}} \frac{\boldsymbol{L(0)}}{\boldsymbol{H}} \int_{t=0}^{\infty} e^{-t[\rho-n-(1-\theta)g]} \frac{c(t)^{1-\theta}}{1-\theta} dt$$

$$= B \int_{t=0}^{\infty} e^{-\beta t} \frac{c(t)^{1-\theta}}{1-\theta} dt \qquad (2.12)$$

Where, $\boldsymbol{B} = \boldsymbol{A(0)^{1-\theta}} \frac{\boldsymbol{L(0)}}{\boldsymbol{H}}$ and $\boldsymbol{\beta} = \boldsymbol{\rho} - \boldsymbol{n} - (\boldsymbol{1} - \boldsymbol{\theta})\boldsymbol{g}$, ($from$ (2.2)β is assumed positive)

Now we consider **the budget constraint**, (2.6):

- Total consumption, $\frac{C(t)L(t)}{H} = \frac{c(t)A(t)L(t)}{H}$
- Total Labor Income, $\frac{W(t)L(t)}{H} = \frac{w(t)A(t)L(t)}{H}$
- Initial capital holdings, $\frac{K(0)}{H} = \frac{k(0)A(0)L(0)}{H}$

Now equation (2.6) can be rewritten as:

$$\int_{t=0}^{\infty} e^{-R(t)} c(t) \frac{A(t)L(t)}{H} dt \leq k(0) \frac{A(0)L(0)}{H} + \int_{t=0}^{\infty} e^{-R(t)} w(t) \frac{A(t)L(t)}{H} dt \qquad (2.13)$$

Here, $A(t)L(t) = A(0)e^{gt}L(0)e^{nt} = A(0)L(0)\boldsymbol{e}^{(\boldsymbol{n}+\boldsymbol{g})\boldsymbol{t}}$. substituting this fact into (2.13) and dividing both sides by $\frac{A(0)L(0)}{H}$ yields:

$$\int_{t=0}^{\infty} e^{-R(t)} c(t) \boldsymbol{e}^{(\boldsymbol{n}+\boldsymbol{g})\boldsymbol{t}} dt \leq k(0) + \int_{t=0}^{\infty} e^{-R(t)} w(t) \boldsymbol{e}^{(\boldsymbol{n}+\boldsymbol{g})\boldsymbol{t}} dt \qquad (2.14)$$

Finally, because K(s) is proportional to $k(s)e^{(n+g)s}$, we can rewrite the no-Ponzi-game version of the budget constraint, (2.10), as

$$\lim_{s\to\infty} e^{-R(s)} k(s) e^{(n+g)s} \geq 0 \qquad (2.15)$$

Household Behavior

The household's problem is to choose the path of c(t) to maximize lifetime utility, (2.12), subject to the budget constraint, (2.14).

We set up the Lagrangian:

$$\mathcal{L} = B\int_0^\infty e^{-\beta t}\frac{c(t)^{1-\theta}}{1-\theta}dt + \lambda\left[k(0) + \int_{t=0}^\infty e^{-R(t)}w(t)e^{(n+g)t}dt - \int_{t=0}^\infty e^{-R(t)}c(t)e^{(n+g)t}dt\right] \tag{2.16}$$

The first-order condition for an individual c(t) is:

$$Be^{-\beta t}\, c(t)^{-\theta} = \lambda e^{-R(t)}e^{(n+g)t} \tag{2.17}$$

The household's behavior is characterized by (2.17) and the budget constraint, (2.14).

Taking logs of both sides of (2.17)

$$\ln B - \beta t - \theta \ln c(t) = \ln\lambda - R(t) + (n+g)t$$

$$= \ln\lambda - \int_{\tau=0}^{t} r(\tau)d\tau + (n+g)t \tag{2.18}$$

[Since, by definition $R(t) = \int_{\tau=0}^{t} r(\tau)d\tau$]

Taking the derivative of both sides of (2.18) with respect to t

$$-\beta - \theta\frac{\dot{c}(t)}{c(t)} = -r(t) + (n+g) \tag{2.19}$$

[Since, the time derivative of the log of a variable equals its growth rate]

Now,

$$\frac{\dot{c}(t)}{c(t)} = \frac{r(t)-n-g-\beta}{\theta}$$

$$\frac{\dot{c}(t)}{c(t)} = \frac{r(t)-\rho-\theta g}{\theta} \qquad = \tag{2.20}$$

[Since, $\beta = \rho - n - (1-\theta)g$]

As, C(t) = c(t)A(t). The the growth rate of C is given by:

$$\frac{\dot{C}(t)}{C(t)} = \frac{\dot{A}(t)}{A(t)} + \frac{\dot{c}(t)}{c(t)}$$

$$= g + \frac{r(t)-\rho-\theta g}{\theta}$$

$$= \frac{r(t)-\rho}{\theta} \tag{2.21}$$

This condition states that consumption per worker is rising if the real return exceeds the rate at which the household discounts future consumption [$r(t) > \rho$], and is falling if $r(t) < \rho$.

The smaller is θ—the less marginal utility changes as consumption changes—the larger are the changes in consumption in response to differences between the real interest rate and the discount rate.

Equation (2.20) is known as the *Euler equation* for this maximization problem.

Intuitively, the Euler equation describes **how c must behave over time given c(0)**: if c does not evolve according to (2.20), the household can rearrange its consumption in a way that raises its lifetime utility without changing the present value of its lifetime spending.

2.3 The Dynamics of the Economy

The Dynamics of *c*

Equation (2.20) describes the evolution of c (**consumption per worker**) for the economy as a whole, since all households are the same.

Since, $\mathbf{r(t) = f'(k(t))}$, we can rewrite (2.20) as

$$\frac{\dot{c}(t)}{c(t)} = \frac{\mathbf{f'(k(t))} - \rho - \theta g}{\theta} \tag{2.24}$$

Thus, $\dot{\boldsymbol{c}}$ is **zero** (at the **BGP**) when $\mathbf{f'(k(t) = \rho + \theta g}$. Let k^* denote this level of k.

When $\boldsymbol{k > k^*}$, $f'(k) < \rho + \theta g$, and thus $\dot{c}$ is **negative**. Conversely, when $\boldsymbol{k <\ k^*}$, $\dot{c}$ is **positive**.

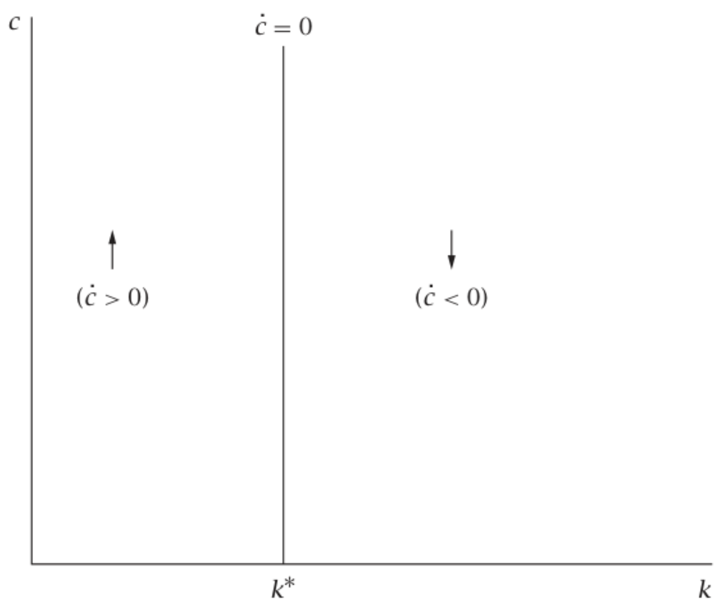

FIGURE 2.1 The dynamics of *c*

This information is visualized in figure 2.1 The arrows show the direction of motion of *c*.

Thus, c is rising if $\mathrm{k} <\ \mathrm{k}^*$, and falling if $\mathrm{k} >\ \mathrm{k}^*$.

The $\dot{c} = 0$ at $k = k^*$ indicates that *c* is constant for this value of k.

The Dynamics of *k*

$\dot{k}$ = actual investment − break-even investment

Actual investment is output minus consumption, i.e., ***f(k) − c***

Since we are assuming that there is **no depreciation**, break-even investment is ***(n + g)k***

That is,

$$\dot{k}(t) = f\big(k(t)\big) - c(t) - (n+g)k(t) \tag{2.25}$$

For a given k, **the level of *c*** that implies $\dot{\boldsymbol{k}} = \mathbf{0}$ is given by $\boldsymbol{f(k) - (n+g)k}$. This value of c increases with k until $\boldsymbol{f'(k) = n + g}$ (**the golden-rule level of *k***) and then decreases. When c exceeds the level that yields $\dot{k} = 0$, k is falling; when c is less than this level, k is rising.

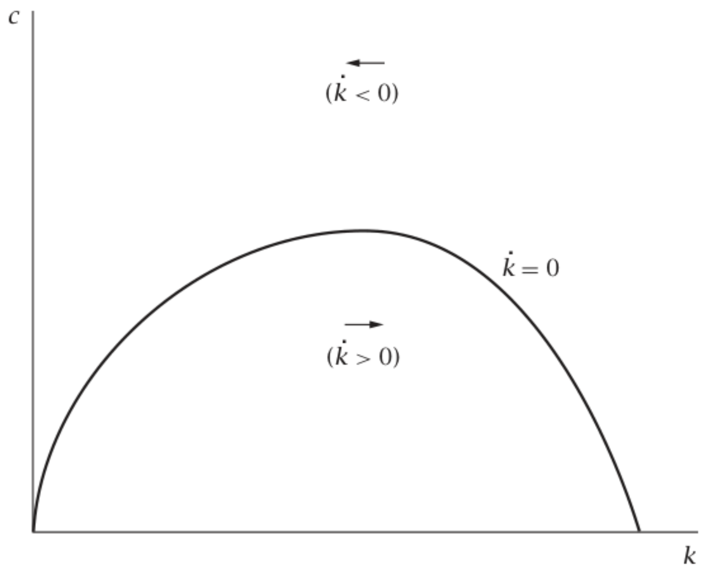

FIGURE 2.2 The dynamics of *k*

This information is summarized in Figure 2.2; the arrows show the direction of motion of k.

The Phase Diagram

Combining the dynamics of c and k provides a complete picture of the economy's behavior. The phase diagram shows the directions of motion of both c and k:

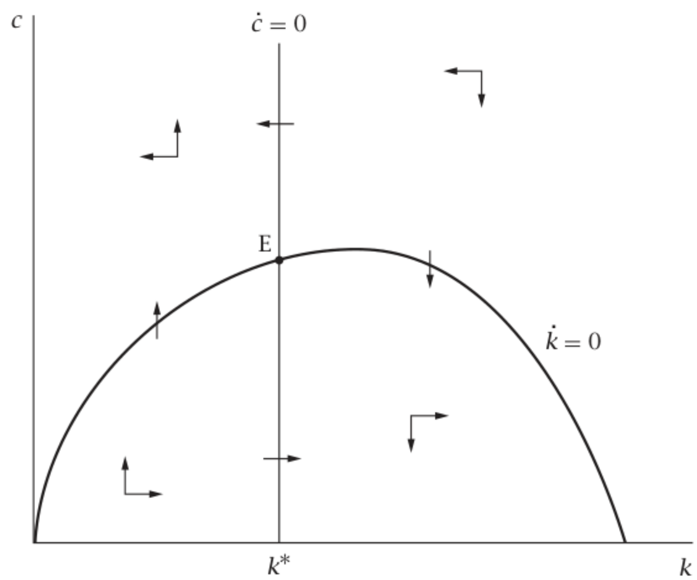

FIGURE 2.3 The dynamics of c and k

For example,

- To the **left of the $\dot{c} = 0$** locus and **above the $\dot{k} = 0$** locus, **$\dot{c}$ is positive** and **$\dot{k}$ is negative**. Thus ***c* is rising** and ***k* falling**, and so the arrows point **up and to the left**.

The arrows in the other sections of the diagram are based on similar reasoning.

- At the equilibrium point (Point E), both **$\dot{c}$ and $\dot{k}$ are zero**; thus, there is **no movement** from this point.

Note that, Figure 2.3 is drawn with k^* **less than** the golden-rule level of k.

The Initial Value of c

Figure 2.3 shows how c and k must **evolve over time** to satisfy equation (2.24) and (2.25) ***given initial values of c and k.***

- The initial value of k is given; but the **initial value of *c* must be determined**.
- This issue is addressed in Figure 2.4
- For concreteness, $k(0)$ is assumed to be **less than k^***
- The figure shows the trajectory of c and k for various assumptions concerning the initial level of c

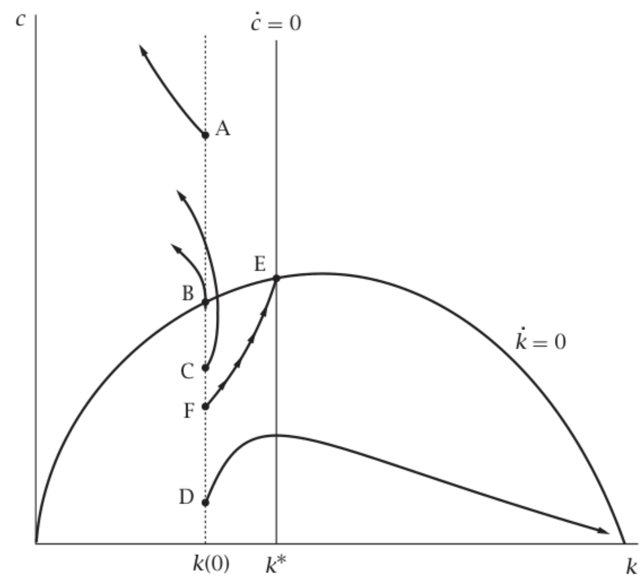

FIGURE 2.4 The behavior of c and k for various initial values of c

- **If c(0) is above the $\dot{k} = 0$** curve, at a **point like A**, then **$\dot{c}$ is positive** and **$\dot{k}$ negative**; thus the **economy** moves continually **up and to the left** in the diagram.

- **If $c(0)$ is such that $\dot{k}$ is initially zero** **(Point B),** the economy begins by moving **directly up** in (k,c) space; thereafter **$\dot{c}$ is positive** and **$\dot{k}$ negative**, and so the **economy** again moves **up and to the left**.

- **If the economy begins slightly below the $\dot{k}$ = 0 locus** **(Point C), $\dot{k}$ is initially positive but small** (since $\dot{k}$ is a continuous function of c), and **$\dot{c}$ is again positive**. Thus in this case the **economy** initially moves **up and slightly to the right**; **after it crosses the $\dot{k}$ = 0** locus, however, **$\dot{k}$ becomes negative** and once again the **economy** is on a path of **rising c** and **falling k.**

- **A case of very low initial consumption** is shown in a **point D**. Here **$\dot{c}$ and $\dot{k}$** are both initially **positive**. From (2.24), $\dot{c}$ is proportional to c; **when c is small, $\dot{c}$ is therefore small**. Thus **c remains low**, and so the economy eventually crosses the $\dot{c}$ = 0 line. **After this point**, **$\dot{c}$ becomes negative**, and **$\dot{k}$ remains positive**. Thus the **economy** moves **down and to the right**.

- **Critical point between Points C and D—Point F:** $\dot{c}$ and $\dot{k}$ are continuous functions of c and k. Thus there is some **critical point** between Points C and D—**Point F** in the diagram—such that at that level of initial c, the economy **converges to the stable point**, **Point E.**

Consumption above the critical level:
The $\dot{k}$ = 0 curve is crossed before the $\dot{c}$ = 0 line is reached,
The economy ends up on a path of **perpetually rising consumption and falling capital**.

Consumption less than the critical level:
The $\dot{c}$ = 0 locus is reached first
The economy embarks on a path of **falling consumption and rising capital**.

Consumption is just equal to the critical level:
The economy converges to the point where both ***c* and *k* are constant.**

The Saddle Path

For any positive initial level of k, there is a unique initial level of c that is consistent with
- households' intertemporal optimization,
- the dynamics of the capital stock,
- households' budget constraint, and
- the requirement that k not be negative.

The function giving this initial *c* as a function of *k* is known as the *saddle path*; it is shown in Figure 2.5.

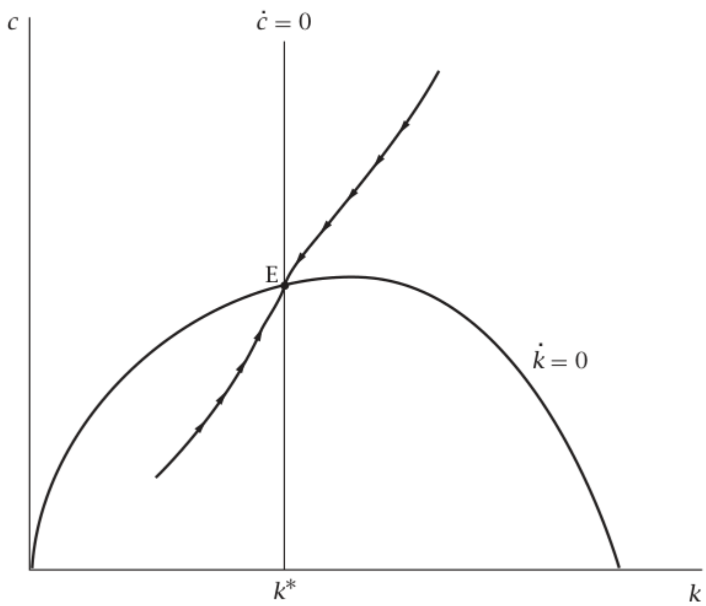

FIGURE 2.5 The saddle path

For any starting value for k, the initial ***c* must be the value on the saddle path**. The economy then moves along **the saddle path** to Point E.

2.4 Welfare

(This part is **not much important** for your exam. Read it for knowledge accumulation. For simplicity, ignore social planner's problem and **read the conclusion**)

Is the equilibrium of this economy a desirable outcome?

According to first welfare theorem: if markets are **competitive** and **complete** and there **are no externalities** (and if the **number of agents is finite**), then the decentralized equilibrium is Pareto-efficient.

Since the conditions of the first welfare theorem hold in RCK model, the equilibrium must be **Pareto-efficient**. And since **all households** have the **same utility**, this means that the **decentralized equilibrium** produces the **highest possible utility** among allocations that treat all households in the same way.

Social planner's problem:

To see this more clearly, consider the problem facing a **social planner** who can dictate the division of output between consumption and investment at each date and who **wants to maximize the lifetime utility of a representative household**. This problem is identical to that of an individual household except that, rather than taking the paths of **w and r** as given, the planner takes into account the fact that **these are determined by the path of *k***, which is in turn determined by (2.25).

$\frac{\dot{c}(t)}{c(t)} = \frac{r(t)-\rho-\theta g}{\theta}$ =	(2.20)
$\frac{\dot{c}(t)}{c(t)} = \frac{f'(k(t))-\rho-\theta g}{\theta}$	(2.24)
$\dot{k}(t) = f(k(t)) - c(t) - (n+g)k(t)$	(2.25)

The intuitive argument:

- consumption at consecutive moments used to derive (2.20) or (2.24) applies to the social planner as well: **reducing c by Δc** at time t and investing the proceeds allows the planner to **increase *c*** at **time $t+\Delta t$** by $e^{f'(k(t))\Delta t}e^{-(n+g)\Delta t}\Delta c$
- Thus $c(t)$ along the path chosen by the planner must satisfy (2.24).
- And since equation (2.25) giving the evolution of k reflects technology, not preferences, the social planner must obey it as well.
- Finally, as with households' optimization problem, paths that require that the capital stock becomes negative can be ruled out on the grounds that they are not feasible, and
- Paths that cause consumption to approach zero can be ruled out on the grounds that they do not maximize households' utility.

Conclusion: the **solution to the social planner's problem** is for the initial value of c to be given by the value on the **saddle path**, and for c and k to then move along the saddle path.

That is, **the competitive equilibrium maximizes the welfare of the representative household.**

2.5 The Balanced Growth Path

Properties of the Balanced Growth Path

Once the economy has converged to point E, the behavior of the economy is identical to that of Solow economy on the BGP.

- Capital, output, and consumption **per unit of effective labor (k,y,c) are constant**
- Since y and c are constant, the saving rate, **s** $= \frac{y-c}{y}$ **is also constant**
- The **total capital** stock, **total output**, and **total consumption** grow at rate $n + g$
- Capital per worker **(K)**, output per worker**(Y),** and consumption per worker**(C)** grow at rate g

Even when saving is endogenous, **growth in the effectiveness of labor** remains the **only source of persistent growth** in output per worker.

The Social Optimum and the Golden-Rule Level of Capital

- The **only notable difference** between the BGP of the Solow and RCK model is that a **BGP with a capital stock above the golden-rule level is not possible in the RCK model.**

- In the **Solow model**, a sufficiently high saving rate causes the economy to reach a BGP with the property that **there are feasible alternatives that involve higher consumption** at every moment.

- In contrast, in the RCK model, **saving** is derived from the behavior of households whose **utility depends on their consumption**, and there are no externalities.

- As a result, **it cannot be an equilibrium** for the economy to follow a path where higher consumption can be attained in every period;

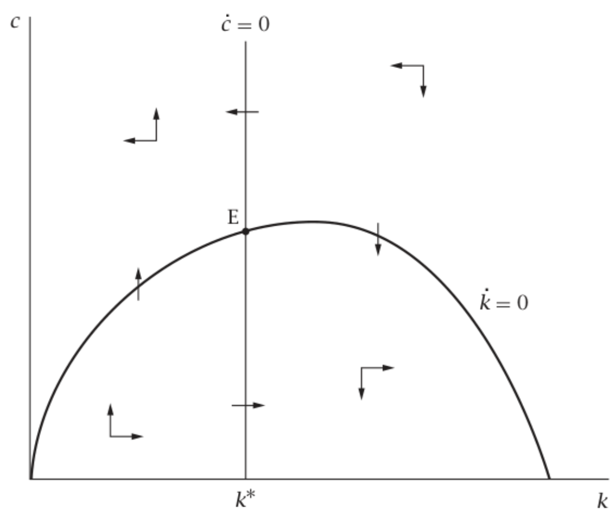

FIGURE 2.3 The dynamics of c and k

From the phase diagram, If the initial capital stock exceeds the golden-rule level, i.e., $k(0) > k_{GR}$ initial consumption is above the level needed to keep k constant; thus $\dot{k}$ **is negative**. k gradually approaches k^*, **which is below the golden-rule level**.

Finally, the fact that k^* is less than the golden-rule capital stock implies that **the economy does not converge to the BGP that yields the maximum sustainable level of *c*.**

(Ignore the following part if you find it difficult)

Intution:

- The intuition for this result is clearest in the case of $g = 0$
- $g = 0$ implies here is no long-run growth of consumption and output per worker.
- In this case, k^* is defined by $f'(k^*) = \rho$ (see [2.24]) and
- k_{GR} is defined by $f'(k_{GR}) = n$
- Our assumption that $\rho - n - (1-\theta)g > 0$ simplifies to $\rho > n$
- $\rho > n$ implies $k^* < k_{GR}$
- Thus an **increase in saving** starting at $k = k^*$would cause consumption per worker **(C)** to eventually **rise above its previous level** and remain there.

- But households value **present consumption** more than future consumption.
- At some point, when k exceeds $k*$— accepting a **temporary short-term sacrifice** in consumption for **permanent long-term gain** **reduces** rather than raises **lifetime utility**.

- **Thus k converges to a value below the golden-rule level**

(Ignore the above part if you find it difficult)

Because k^***is the optimal level of *k*** for the economy to converge to, it is known as **the *modified golden-rule* capital stock.**

2.6 The Effects of a Fall in the Discount Rate

Consider a RCK economy on its BGP and there is a **fall in ρ**

Division of output between consumption and investment is determined by **forward-looking households** We must specify whether the change is **expected or unexpected**. If a change is expected, households may alter their behavior before the change occurs.

We therefore focus on the simple case where the **change is unexpected**

Qualitative Effects

- The **evolution of k** is determined by **technology** rather than preferences,
- ρ enters the **equation for $\dot{c}$** but not the one for $\dot{k}$.
- Thus **only the $\dot{c} = 0$ locus is affected.**

Equation (24): $\frac{\dot{c}(t)}{c(t)} = \frac{f'(k(t)) - \rho - \theta g}{\theta}$

- Thus the value of k where **$\dot{c}$ equals zero** is defined by $\mathbf{f'(k^*) = \rho + \theta g}$
- Since $f''(\cdot)$ **is negative**, this means that the **fall in ρ raises k^***
- **Thus the $\dot{c} = 0$ line shifts to the right.**

This is shown in Figure 2.6.

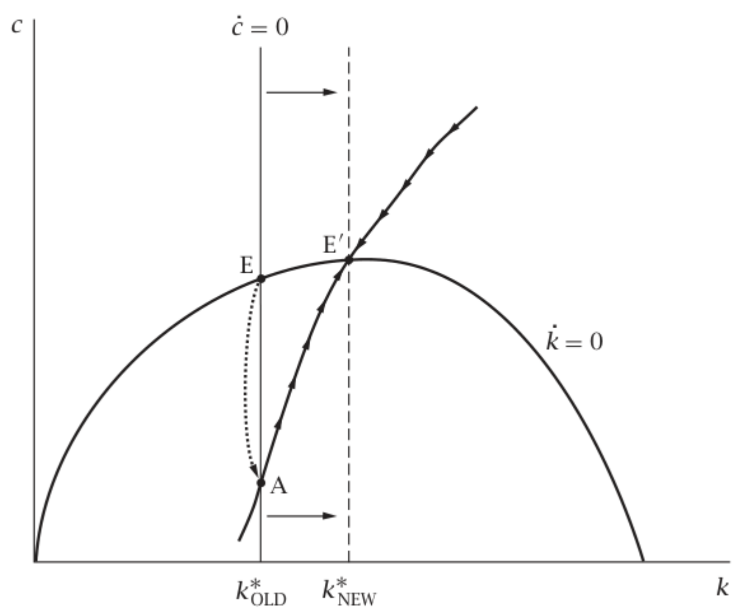

FIGURE 2.6 The effects of a fall in the discount rate

- Since k is given by the history of the economy, it **cannot change discontinuously,** at the time of the change in ρ
- $k = k^*$ on the old BGP.
- In contrast, **c can jump** at the time of the shock

Therefore, when **ρ falls**

- At the instant of the change, **c jumps down** so that **the economy** is **on the new saddle path** (Point A in Figure 2.6).
- Thereafter, **c and k rise gradually** to their **new BGP values**, these are higher than their values on the original BGP.

Thus, the **permanent fall in the discount rate** produces **temporary increases** in the **growth rates** of **K & Y** (capital per worker and output per worker).

The Rate of Adjustment and the Slope of the Saddle Path

- We replace (2.24) and (2.25) with their **linear approximation** around the BGP.
- Thus we take first-order Taylor approximations to (2.24) and (2.25) around $\boldsymbol{k = k^*, c = c^*}$
- That is, we write

$$\dot{c} \simeq \frac{\partial \dot{c}}{\partial k}[k - k^*] + \frac{\partial \dot{c}}{\partial c}[c - c^*] \tag{2.26}$$

$$\dot{k} \simeq \frac{\partial \dot{k}}{\partial k}\left[k - k^*\right] + \frac{\partial \dot{k}}{\partial c}[c - c^*] \tag{2.27}$$

- Define, $\boldsymbol{\tilde{c} = c - c^*}$ and $\boldsymbol{\tilde{k} = k - k^*}$
- Since c^* and $k^*$are both constant $\boldsymbol{\dot{\tilde{c}} = \dot{c}}$ and $\boldsymbol{\dot{\tilde{k}} = \dot{k}}$
- Therefore (2.26) and (2.27) can be rewritten as:

$$\dot{\tilde{c}} \simeq \frac{\partial \,\dot{c}}{\partial k}\,\tilde{k} + \frac{\partial \dot{c}}{\partial c}\,\tilde{c} \tag{2.28}$$

$$\dot{\tilde{k}} \simeq \frac{\partial \,\dot{k}}{\partial k}\,\tilde{k} + \frac{\partial \dot{k}}{\partial c}\,\tilde{c} \tag{2.29}$$

Using $\boldsymbol{\dot{c} = \frac{[f'(k) - \rho - \theta g]}{\theta} c}$ (from 2.24) to compute the derivatives in (2.28) we get

$$\dot{\tilde{c}} \simeq \frac{\partial}{\partial k}\frac{[f'(k) - \rho - \theta g]}{\theta} c\,\tilde{k} + \frac{\partial}{\partial c}\frac{[f'(k) - \rho - \theta g]}{\theta} c\,\tilde{c}$$

$$\boldsymbol{\dot{\tilde{c}} \simeq \frac{f''(k^*)c^*}{\theta}\tilde{k}} \tag{2.30}$$

[I did not understand how (2.30) is derived. Please let me know if you know the process.]

Using $\boldsymbol{\dot{k} = [f(k) - c - (n+g)k]}$ to find the derivatives in (2.29) we get

$$\dot{\tilde{k}} \simeq \frac{\partial}{\partial k}[f(k) - c - (n+g)k]\,\tilde{k} + \frac{\partial}{\partial c}[f(k) - c - (n+g)k]\,\tilde{c}$$

$$\dot{\tilde{k}} \simeq [f'(k^*) - (n+g)]\,\tilde{k} - \tilde{c}$$

$$\dot{\tilde{k}} \simeq [\rho + \theta g - (n+g)]\,\tilde{k} - \tilde{c} \qquad [on\ BGP, \boldsymbol{f'(k(t) = \rho + \theta g}]$$

$$\dot{\tilde{k}} \simeq [\rho - n - (1-\theta)g]\,\tilde{k} - \tilde{c}$$

$$\boldsymbol{\dot{\tilde{k}} \simeq \beta\,\tilde{k} - \tilde{c}} \tag{2.31}$$

- Dividing both sides of **(2.30) by $\tilde{c}$** and both sides of **(2.31) by $\tilde{k}$** yields expressions for the growth rates of $\tilde{c}$ and $\tilde{k}$

$$\frac{\dot{\tilde{c}}}{\tilde{c}} \simeq \frac{f''(k^*)c^*}{\theta}\frac{\tilde{k}}{\tilde{c}} \tag{2.32}$$

$$\frac{\dot{\tilde{k}}}{\tilde{k}} \simeq \beta - \frac{\tilde{c}}{\tilde{k}} \tag{2.33}$$

- Equations (2.32) and (2.33) imply that the growth rates of $\tilde{c}$ and $\tilde{k}$ depend only on the **ratio of $\tilde{c}$ and $\tilde{k}$.**

- if the values of $\tilde{c}$ and $\tilde{k}$ are falling at the same rate (that is, if they imply ($\frac{\dot{\tilde{c}}}{\tilde{c}} = \frac{\dot{\tilde{k}}}{\tilde{k}}$), this implies that the ratio of $\tilde{c}$ to $\tilde{k}$is not changing, and thus that their growth rates are also not changing.

- That is, if $c - c^*$ and $k - k^*$ are **initially falling** at the same rate, **they continue to fall** at that rate.

- **<u>In terms of the diagram</u>**, from a point where $\tilde{c}$ and $\tilde{k}$ are **falling** at **equal rates**, the economy moves along a straight line to (k^*, c^*), with the **distance** from (k^*, c^*) **falling** at a **constant rate**.

Let μ denote $\frac{\dot{\tilde{c}}}{\tilde{c}}$,Equation (2.32) implies

$$\frac{\tilde{c}}{\tilde{k}} = \frac{f''(k^*)c^*}{\theta}\frac{1}{\mu} \tag{2.34}$$

From (2.33), the condition that $\frac{\dot{\tilde{c}}}{\tilde{c}} = \frac{\dot{\tilde{k}}}{\tilde{k}}$ is thus

$$\mu = \beta - \frac{f''(k^*)c^*}{\theta}\frac{1}{\mu} \tag{2.35}$$

$$\mu^2 - \beta\mu + \frac{f''(k^*)c^*}{\theta} = 0 \tag{2.36}$$

This is a quadratic equation in μ. The solutions are

$$\mu = \frac{\beta \pm \sqrt{\beta^2 - 4\frac{f''(k^*)c^*}{\theta}}}{2} \tag{2.37}$$

Let μ_1 ***and*** μ_2 denote these two values of μ.

- **<u>If μ is positive</u>**, then $\tilde{c}$ and $\tilde{k}$ are **growing**;
- That is, instead of moving along a straight line toward $(\boldsymbol{k}^*, \boldsymbol{c}^*)$, **the economy** is moving on a straight line **away from $(\boldsymbol{k}^*, \boldsymbol{c}^*)$**
- Thus **if the economy is to converge** to $(\boldsymbol{k}^*, \boldsymbol{c}^*)$ then **<u>μ must be negative</u>**.

- Inspection of (2.37) shows that only one of the μ's, namely $\frac{\beta-\sqrt{\beta^2-4\frac{f''(k^*)c^*}{\theta}}}{2}$, **is negative.**
- Let μ_1 denote this value of μ.

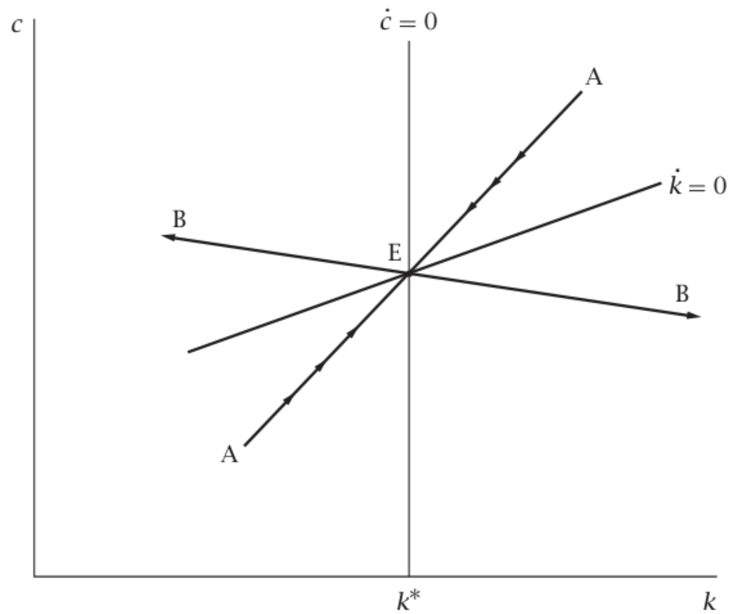

FIGURE 2.7 The linearized phase diagram

- Figure 2.7 shows the line along which the economy **converges** smoothly to $(\boldsymbol{k}^*, \boldsymbol{c}^*)$; it is **labeled AA**
- **AA** is the **saddle path** of the linearized system
- The figure also shows the line along which the economy moves directly away from $(\boldsymbol{k}^*, \boldsymbol{c}^*)$; it is labeled BB
- If $c(0)$ and $k(0)$ lay along this line, $\tilde{c}$ and $\tilde{k}$ would grow steadily at rate μ_2
- Since $\boldsymbol{f''}$ **(•)** is negative, (2.34) implies that the relation between **$\tilde{c}$ and $\tilde{k}$** has the **opposite sign from μ.**
- Thus the saddle path **AA is positively sloped**, and the **BB line is negatively sloped**.

The Speed of Adjustment

Implications of (2.37) for the **speed of convergence** to the BGP:

- In Cobb–Douglas production function $\boldsymbol{f(k) = k^{\alpha}}$
- This implies $\boldsymbol{f''(k^*) = \alpha(\alpha-1){k^*}^{\alpha-2}}$
- consumption per unit of effective labor, $\boldsymbol{c^* = {k^*}^{\alpha} - (n+g)k^*}$
- Thus the **expression for $\boldsymbol{\mu_1}$** can be written as

$$\mu_1 = \frac{1}{2}\left[\beta - \sqrt{\beta^2 - \frac{4}{\theta}\alpha(\alpha-1){k^*}^{\alpha-2}\left[{k^*}^{\alpha} - (n+g)k^*\right]}\right] \tag{2.38}$$

On the BGP, $\boldsymbol{f'(k(t) = \rho + \theta g}$ [from (2.24)]

➢ For the Cobb–Douglas case, this is equivalent to

$$\alpha {k^*}^{\alpha-1} = \rho + \theta g$$

$$k^* = \left(\frac{\boldsymbol{\rho+\theta g}}{\boldsymbol{\alpha}}\right)^{\frac{\mathbf{1}}{\boldsymbol{\alpha-1}}}$$

- Subsituting $k^* = \left(\frac{\boldsymbol{\rho+\theta g}}{\boldsymbol{\alpha}}\right)^{\frac{\mathbf{1}}{\boldsymbol{\alpha-1}}}$ into (2.38) and doing some uninteresting algebraic manipulations yields

$$\mu_1 = \frac{1}{2}\left[\beta - \sqrt{\beta^2 + \frac{4}{\theta}\frac{1-\alpha}{\alpha}(\rho+\theta g)\big(\rho+\theta g - \alpha(n+g)\big)}\right] \tag{2.39}$$

Equation (2.39) expresses **the rate of adjustment** in terms of the underlying parameters of the model

- Suppose, $\alpha = \frac{1}{3},\ \rho = 4\%,\ n = 2\%,\ g = 1\%,\ \theta = 1$

It can be shown that, **on the BGP**

➢ r(t) = 5%
➢ s = 20%
➢ $\beta = \rho - n - (1-\theta)g = 2\%$
➢ $\boldsymbol{\mu_1 \simeq -5.4\%}$.

Thus, the adjustment is quite rapid.

For same values of $\boldsymbol{\alpha, n, g}$: In the **Solow model**, the **adjustment speed** was **2%** (per year)

The reason for the difference:

In this example:

- $\boldsymbol{s > s^*}$ when $\boldsymbol{k < k^*}$
- $\boldsymbol{s < s^*}$ when $\boldsymbol{k > k^*}$

But, **saving rate(s)** was **constant** in the Solow model.

2.7 The Effects of Government Purchases

Adding Government to the Model

- Assume that the government **buys output** at rate ***G(t)*** **per unit of effective labor** per unit time.
- The purchases are **financed by** lump-sum taxes of amount $G(t)$ **per unit of effective labor** per unit time

- Thus the government always runs a **balanced budget**
- Investment is now the **difference between output** and the sum of **private consumption** and **government purchases**
- Thus the equation of motion for k, (2.25), becomes

$$\dot{k}(t) = f\big(k(t)\big) - c(t) - G(t) - (n+g)k(t) \tag{2.40}$$

- **A higher value of *G* shifts the $\boldsymbol{\dot{k} = 0}$ locus down**

- By assumption, households' preferences ([2.1]–[2.2] or [2.12]) are unchanged.
- Thus the **Euler equation** ([2.20] or [2.24]) **continues to hold** as before.
- However, the taxes that finance the government's purchases affect households' budget constraint.

Specifically, the budget constraint (2.14) becomes

$$\int_0^\infty e^{-R(t)}\, c(t)e^{(n+g)t}dt \le k(0) + \int_0^\infty e^{-R(t)}[w(t) - G(t)]e^{(n+g)t}dt \tag{2.41}$$

The Effects of Permanent and Temporary Changes in Government Purchases

Permanent increase in G

- Suppose that the economy is on a BGP with $G(t)$ constant at some level G_L, and that there is an unexpected, permanent increase in G to G_H.
- According to (2.40) The $\boldsymbol{\dot{k} = 0}$ **locus shifts down by the amount of the increase in *G***
- Since government purchases **do not affect the Euler equation**, **the $\boldsymbol{\dot{c} = 0}$ locus remains unaffected**.
- In response to such a change, ***c*** **must jump** so that the **economy is on its new saddle path**.

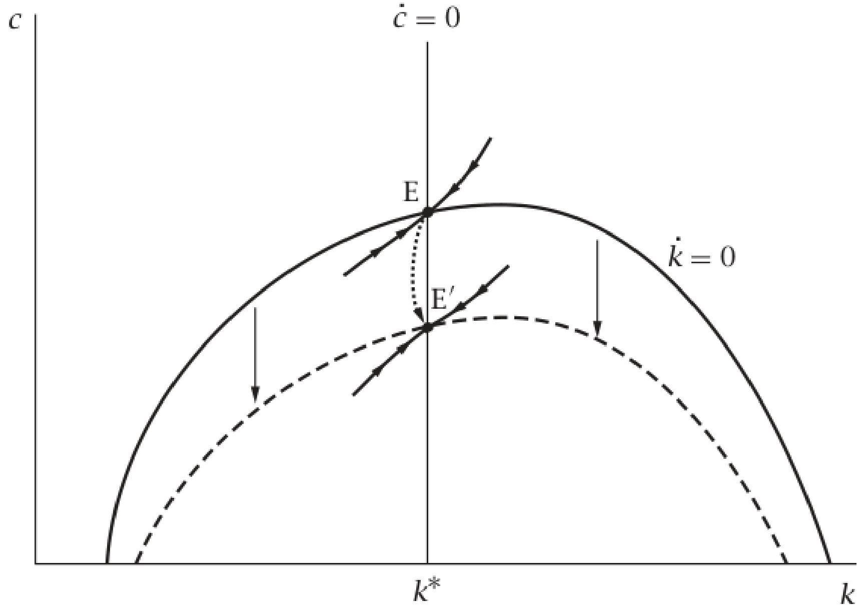

FIGURE 2.8 The effects of a permanent increase in government purchases

The adjustment takes a simple form: ***c*** **falls** by the amount of the increase in *G*, and **the economy** is immediately on its **new BGP.**

(Ignore following part for exam)

Intuitively,

- The permanent increases in government purchases and taxes reduce households' lifetime wealth.
- And there is **no scope** for households to **raise their utility by adjusting the time pattern of their consumption.**
- Thus the **size of the immediate fall in consumption** is **equal to the full amount of the increase in government purchases.**
- And the **capital stock** and the **real interest rate** are **unaffected** since a **permanent increase** in government purchases **does not cause crowding out** of investment.

(Ignore above part for exam)

Temporary increase in G

- There is an unanticipated increase in *G* which is expected to be temporary.
- In this case, *c* does not fall by the full amount of the increase in *G*, $G_H - G_L$
- **At the time of the increase in *G*:** ***c* must jump** to the value such that the dynamics implied by (2.40) with $G = G_H$ (and by [2.24]) bring the economy **to the old saddle** path at the time that ***G* returns to its initial level.**

- Thereafter, the economy moves along that saddle path to the old BGP.

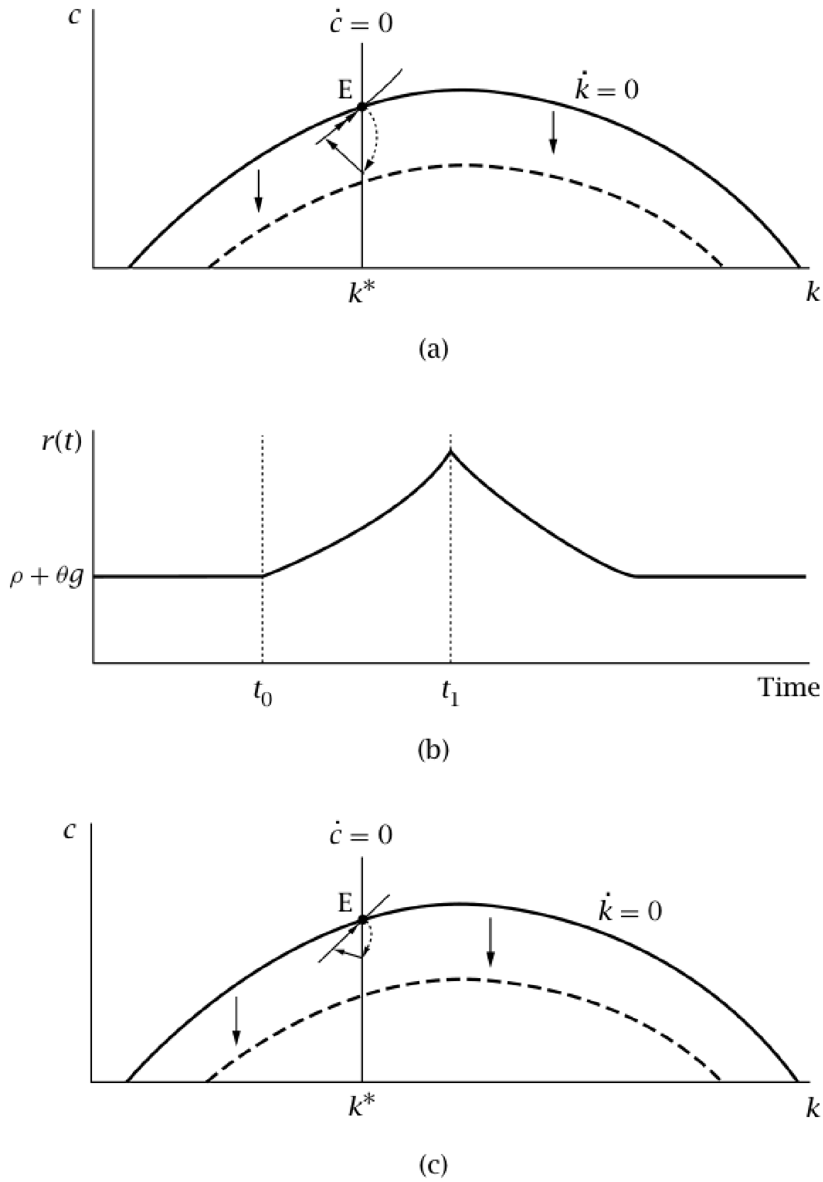

FIGURE 2.9 The effects of a temporary increase in government purchases

Panel (a) shows a case where the increase in *G* is relatively long-lasting:

- In this case ***c* falls by most of the amount of the increase in *G*.**
- Households somewhat decrease their capital holdings.
- ***c* rises** as the economy approaches the time that ***G* returns to G_L**

Panel (b) shows the behavior of r when the increase in *G* is relatively long-lasting

- Since $\boldsymbol{r} = \boldsymbol{f}'(\boldsymbol{k})$, we can deduce the **behavior of *r*** from the behavior of *k*.
- Thus ***r* rises gradually** during the period that **government spending is high**.
- And then **gradually returns to its initial level**.

t_0 = time of increase in G.
t_1 = time of its return to its initial value.

Panel (c) shows the case of a short-lived rise in *G*

- In this case, **households reduce their consumption relatively little**.
- Instead, household choose to **pay** for most of their **temporary higher taxes** out of their **savings**.
- Because government purchases are high for only a short period, the effects on the **capital stock** and the **real interest rate are small.**

Empirical Application: Wars and Real Interest Rates

- This analysis suggests that **temporarily high government purchases cause real interest rates to rise**.
- Whereas **permanently high purchases do not.**

A natural example of a period of **temporarily high government purchases** is **a war**.

Thus our analysis predicts that **real interest rates are high** during **wars.**

- **Barro (1987)** tests this prediction by examining **military spending** and **interest rates** in the United Kingdom from 1729 to 1918.
- **Barro (1987)** found there is a **statistically significant link** between **temporary military spending** and **interest rates.**

However, the success of the theory is not universal.

- In particular, **the real interest rates** of the United States generally appear to be **lower during wars** than in other periods

The reasons for this anomalous behavior are **not well understood**.

Thus the theory **does not provide a full account** of how real interest rates respond to changes in government purchases.